

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

**Cleaning solution**
**Cleaning solution**

CSRCSCS1LI; XXCSRCS1LI; NC2039937

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                   |
|------------------------------------|-------------------|
| Flammable liquids                  | Category 2 (H225) |
| Acute oral toxicity                | Category 4 (H302) |
| Acute dermal toxicity              | Category 4 (H312) |
| Acute Inhalation Toxicity - Vapors | Category 4 (H332) |
| Serious Eye Damage/Eye Irritation  | Category 2 (H319) |

**Label Elements**


Signal Word

Danger

Hazard Statements

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H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation  
H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxicity to Soil Dwelling Organisms  
Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component    | CAS No    | Weight % |
|--------------|-----------|----------|
| Acetonitrile | 75-05-8   | 95       |
| Water        | 7732-18-5 | 4        |
| Formic acid  | 64-18-6   | 1        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water.

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**Inhalation** Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Special hazards arising from the substance or mixture**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Hazardous Combustion Products**

None under normal use conditions.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental precautions**

Should not be released into the environment.

**Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Take precautionary measures against static discharges. Use spark-proof tools and explosion-proof equipment.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

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## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

### Conditions for Safe Storage, Including any Incompatibilities

Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component    | Malaysia | ACGIH TLV                  | OSHA PEL                                                                                                                                                                                                          |
|--------------|----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile |          | TWA: 20 ppm<br>Skin        | (Vacated) TWA: 40 ppm<br>(Vacated) TWA: 70 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 60 ppm<br>(Vacated) STEL: 105 mg/m <sup>3</sup><br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> |
| Formic acid  |          | TWA: 5 ppm<br>STEL: 10 ppm | (Vacated) TWA: 5 ppm<br>(Vacated) TWA: 9 mg/m <sup>3</sup><br>TWA: 5 ppm<br>TWA: 9 mg/m <sup>3</sup>                                                                                                              |

| Component    | European Union                                               | The United Kingdom                                                                                               | Germany                                                                                                                                                                                                                                                                                                                                      |
|--------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 40 ppm (8hr)<br>TWA: 70 mg/m <sup>3</sup> (8hr)<br>Skin | STEL: 60 ppm 15 min<br>STEL: 102 mg/m <sup>3</sup> 15 min<br>TWA: 40 ppm 8 hr<br>TWA: 68 mg/m <sup>3</sup> 8 hr  | TWA: 10 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 17 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK<br>TWA: 17 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 34 mg/m <sup>3</sup> Höhepunkt: 2 mg/m <sup>3</sup><br>Haut |
| Formic acid  | TWA: 5 ppm 8 hr<br>TWA: 9 mg/m <sup>3</sup> 8 hr             | STEL: 15 ppm 15 min<br>STEL: 28.8 mg/m <sup>3</sup> 15 min<br>TWA: 5 ppm 8 hr<br>TWA: 9.6 mg/m <sup>3</sup> 8 hr | TWA: 5 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 9.5 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 5 ppm (8 Stunden). MAK<br>TWA: 9.5 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 10 ppm<br>Höhepunkt: 19 mg/m <sup>3</sup>                                                                                     |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

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## Personal protective equipment

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>   | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                                                                                                                           |
| <b>Recommended Filter type:</b> | low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| <b><u>Hygiene Measures</u></b> | Handle in accordance with good industrial hygiene and safety practice |
|--------------------------------|-----------------------------------------------------------------------|

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b><u>Environmental exposure controls</u></b> | No information available |
|-----------------------------------------------|--------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                          |                                          |
|-------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                   |                          |                                          |
| <b>Physical State</b>               | Liquid                   |                                          |
| <b>Odor</b>                         | No information available |                                          |
| <b>Odor Threshold</b>               | No data available        |                                          |
| <b>pH</b>                           | No information available |                                          |
| <b>Melting Point/Range</b>          | No data available        |                                          |
| <b>Softening Point</b>              | No data available        |                                          |
| <b>Boiling Point/Range</b>          | No information available |                                          |
| <b>Flash Point</b>                  | 2 °C / 35.6 °F           | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | No data available        |                                          |
| <b>Flammability (solid,gas)</b>     | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>             | No data available        |                                          |
| <b>Vapor Pressure</b>               | No data available        |                                          |
| <b>Vapor Density</b>                | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>   | No data available        |                                          |
| <b>Bulk Density</b>                 | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>             | No information available |                                          |
| <b>Solubility in other solvents</b> | No information available |                                          |

**Partition Coefficient (n-octanol/water)**

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|                                  |                          |                                             |
|----------------------------------|--------------------------|---------------------------------------------|
| <b>Component</b>                 | <b>log Pow</b>           |                                             |
| Acetonitrile                     | -0.34                    |                                             |
| Formic acid                      | -0.54                    |                                             |
| <b>Autoignition Temperature</b>  | No data available        |                                             |
| <b>Decomposition Temperature</b> | No data available        |                                             |
| <b>Viscosity</b>                 | No data available        |                                             |
| <b>Explosive Properties</b>      |                          | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>      | No information available |                                             |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                                 |                               |
|---------------------------------|-------------------------------|
| <b>Hazardous Polymerization</b> | No information available.     |
| <b>Hazardous Reactions</b>      | None under normal processing. |

### Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|                   |            |
|-------------------|------------|
| <b>Oral</b>       | Category 4 |
| <b>Dermal</b>     | Category 4 |
| <b>Inhalation</b> | Category 4 |

### Toxicology data for the components

| Component    | LD50 Oral           | LD50 Dermal             | LC50 Inhalation              |
|--------------|---------------------|-------------------------|------------------------------|
| Acetonitrile | 450-787 mg/kg (Rat) | > 2000 mg/kg ( Rabbit ) | LC50 = 3587 ppm (6.022 mg/l) |

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|             |                    |   |                                                         |
|-------------|--------------------|---|---------------------------------------------------------|
|             | 2460 mg/kg ( Rat ) |   | (Mouse) 4h<br>LC50 = 16,000 ppm (26.8 mg/l)<br>(Rat) 4h |
| Water       | -                  | - | -                                                       |
| Formic acid | 730 mg/kg ( Rat )  | - | 7.85 mg/l (Rat) 4h OECD 403                             |

| Component    | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|--------------|-----------------------|-------------------------|-----------------------------|
| Acetonitrile | ATE = 617 mg/kg       | -                       | -                           |

ECHA (RAC) - Committee for Risk Assessment - European CHemicals Agency  
ATE - Acute Toxicity Estimate; mg/kg bw - milligrams per kilogram of body weight

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component    | Freshwater Fish                                                                                                                                                              | Water Flea | Freshwater Algae | Microtox                                                               |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|------------------------------------------------------------------------|
| Acetonitrile | LC50: = 1850 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 1000 mg/L, 96h static (Pimephales promelas)<br>LC50: 1600 - 1690 mg/L, 96h flow-through (Pimephales promelas) |            |                  | EC50 = 28000 mg/L 48 h<br>EC50 = 73 mg/L 24 h<br>EC50 = 7500 mg/L 15 h |

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|             |                                                     |                    |                    |                      |
|-------------|-----------------------------------------------------|--------------------|--------------------|----------------------|
|             | LC50: = 1650 mg/L, 96h static (Poecilia reticulata) |                    |                    |                      |
| Formic acid | Leuciscus idus: LC50 = 46-100 mg/L/96h              | EC50 = 34 mg/L/48h | EC50 = 25 mg/L/96h | EC50 = 46.7 mg/L/17h |

**Persistence and degradability**  
**Persistence** No information available  
Persistence is unlikely.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Acetonitrile | -0.34   | No data available             |
| Formic acid  | -0.54   | 0.22 dimensionless            |

**Mobility in soil** No information available.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

| Component   | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|-------------|------------------------------------------|--------------------------------------------------|
| Formic acid | Applicable                               |                                                  |

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with local regulations

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1648  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ACETONITRILE

### Road and Rail Transport

UN-No UN1648  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ACETONITRILE

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## IATA

UN-No UN1648  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ACETONITRILE

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component    | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Acetonitrile | 200-835-2 | X    | X   | X     | X    | X    | X     | X    | KE-00067 |
| Water        | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |
| Formic acid  | 200-579-1 | X    | X   | X     | X    | X    | X     | X    | X        |

| Component   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Formic acid |                                                                                           |                                                                                          |                            | Annex I - Y34                      |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

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## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date 23-Mar-2025  
Revision Summary Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health  
(Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**